

Experiment 06: Interfacing PIR Motion Sensor with Arduino UNO

Passive Infrared Motion Detection & Alert Telemetry — Program Source Code & Experimental Output

1. Arduino C++ Source Code

Listing 6.1: Microcontroller Firmware for PIR Motion Sensing and Serial Event Telemetry.

```
1 /*
2  * Experiment 06: PIR Motion Sensor Interfacing with Arduino UNO
3  * Hardware Pin Connections:
4  *   - PIR Sensor OUT (O/P) -> Arduino Digital Pin 8 (Configured as INPUT)
5  *   - PIR Sensor VCC       -> Arduino 5V Power Rail
6  *   - PIR Sensor GND       -> Arduino Common Ground (GND)
7  * Serial Baud Rate: 9600 bps | Sampling Interval: 100 ms
8  */
9
10 const int motionPin = 8; // Digital input pin connected to PIR OUT
11
12 void setup() {
13   pinMode(motionPin, INPUT); // Configure Pin 8 as digital input
14   Serial.begin(9600);        // Initialize UART Serial Interface at 9600 Baud
15 }
16
17 void loop() {
18   // Acquire digital logic state from PIR pyroelectric sensor module
19   if (digitalRead(motionPin) == HIGH) {
20     Serial.println("motion detected!");
21   } else {
22     Serial.println("No motion");
23   }
24   delay(100); // 100 ms sampling cycle delay to prevent buffer saturation
25 }
```

2. Experimental Output, Detection State Log & Digital Waveforms

A. Motion Detection Observation Log:

#	Human Target Motion State	Pin 8 Output	Serial Output	Hardware Alert Status
1	Quiescent / Idle Baseline	LOW (0 V)	No motion	Quiescent Baseline (Sensor Armed)
2	Active Hand Movement	HIGH (5 V)	motion det.!	Motion Triggered (T_{hold} Active)
3	Continuous Motion Transit	HIGH (5 V)	motion det.!	Retrigger Interval Extended
4	Target Stationary State	LOW (0 V)	No motion	Quiescent Baseline (Reset)
5	Secondary Transit Cycle	HIGH (5 V)	motion det.!	Motion Triggered (T_{hold} Active)

B. Serial Monitor Stream (9600 Baud):

```
motion detected!
motion detected!
No motion
No motion
motion detected!
```

C. PIR Sensor Thermal Flux & Digital Output Timing Waveform (T_{hold}):

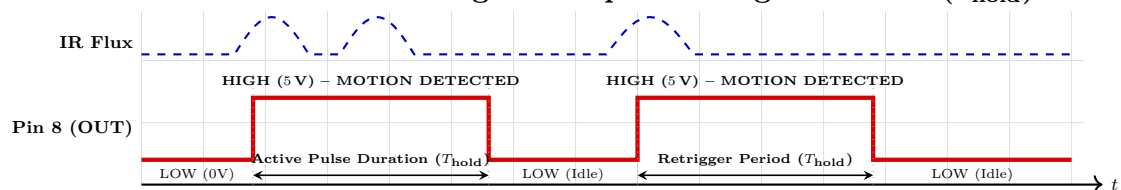


Figure 6.1: PIR sensor analog IR flux response and digital retrigger timing waveform (T_{hold}).